# Amortized analysis exercises

## 17-E01 Aggregate table cost

Difficulty: Basic

A dynamic array starts with capacity1 and doubles when full. For16 pushes, find total insertion/copy cost and average per operation.

## 17-E02 Accounting credit

Difficulty: Advanced

Charge3 per insertion in a capacity-one doubling table. Track credit through capacity4 and verify that growth is paid for.

## 17-E03 Shrink thrashing

Difficulty: Research

A capacity-eight table at size4 immediately shrinks to4. What happens on the next insertion? Give a strategy avoiding thrashing and explain the cost.

## 17-E04 Counter potential with a nonzero initial state

Difficulty: Advanced

A b-bit all-ones counter increments once modulo2^b. With flip cost1 and $\Phi$=number of one bits, find actual/amortized cost and a bound for any m increments.

## 17-E05 Mixed resize potential cases

Difficulty: Research

Start empty,insert5,and delete5. Double when full,halve when post-deletion nonempty $n \leq M/4$,and release when empty. Cost is1 plus copied items. With $\Phi=2n-M$ above half and $M/2-n$ below half,list $c,\Phi,\hat{c}$ for all ten operations and check sums.

## 17-E06 Bit cost of a growing counter

Difficulty: Research

A counter increments from0 and adds a highest bit when needed. Find the exact total flips through value m. If DECREMENT is also available,do arbitrary mixed increment/decrement sequences retain constant amortized flip cost?

> Complete your derivations before opening the separate solutions PDF.

# Amortized analysis exercises (continued)

### References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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